

The Size of Sets

The Hausdorff Measure and Dimension

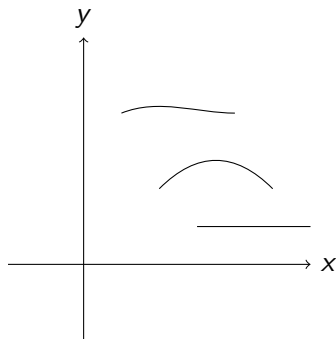
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Outline

- The Lebesgue measure works well for common sets, such as intervals, squares, cubes
- But how do we measure sets with complicated geometry? We need a new way to measure the size of these complicated sets, and assign them dimensions
- Examples:
 - Fractals
 - Sets with non-natural dimensionality
 - The Cantor Set is an example of both
- The Lebesgue measure has limitations
 - Take the following example

Motivation

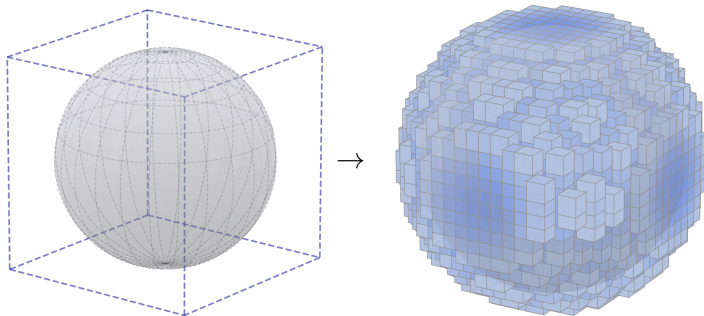


- Let A be this curve. This curve has no area, so $\lambda_2(A) = 0$. This is because the dimension of this curve is not 2. On the other hand, $\lambda_1(A)$ is not defined. So, how can we measure this set? We use the Hausdorff measure

Intuitive Understanding

The idea is if A is n -dimensional, $\text{diam } A$ represents the longest side length of A so $(\text{diam } A)^n$ is the volume of an n -dimensional cube covering A .

- 1 If we cover A with a covering $\{E_i\}_{i \in \mathbb{N}}$, then $\sum_{i=1}^{\infty} \text{diam}(E_i)^n$ is the sum of all n -dimensional cubes covering each E_i and thus also A .
- 2 As the length of the cubes gets smaller, this sum converges to the volume of A . This accounts for "roughness" in a set



Hausdorff Measure: Definition

Definition (Hausdorff d -measure)

Let (X, d) be a metric space with $\text{diam}(\emptyset) := 0$. For $d \geq 0$ and $\delta > 0$, define:

$$\mathcal{H}_\delta^d(A) := \inf \left\{ \sum_{i=1}^{\infty} (\text{diam}(E_i))^d : A \subseteq \bigcup_{i=1}^{\infty} E_i \text{ and } \text{diam}(E_i) \leq \delta \right\} \quad (1)$$

If A does not possess any covering, we have $\mathcal{H}_\delta^d(A) := \inf \emptyset = \infty$.

The d -dimensional Hausdorff measure is then:

$$\mathcal{H}^d(A) := \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^d(A) = \sup_{\delta > 0} \mathcal{H}_\delta^d(A) \quad (2)$$

If $d = 0$: Define $\mathcal{H}^0(\emptyset) = 0$ and use the interpretation $0^0 = 1$ for any other sets so $\mathcal{H}^0$ is the counting measure.

- All we require is $\text{diam}(E_i) \leq \delta$ for each $i \in \mathbb{N}$, so if $\delta \leq \delta'$, then $\mathcal{H}_\delta^d(A) \geq \mathcal{H}_{\delta'}^d(A)$. The supremum form above is thus valid
- An alternative definition allows us to normalise the Hausdorff measure:
 $\mathcal{H}^d(A) = c_d \cdot \lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^d(A)$, but we will use (2)

Hausdorff Measure: Properties

For the d -dimensional Hausdorff measure $\mathcal{H}^d$:

- It is Borel measure (all Borel sets are Hausdorff measurable)
- Connection to Lebesgue: If $c_d = \frac{\pi^{d/2}}{2^d \Gamma(d/2+1)}$ then $\mathcal{H}^d$ equals the Lebesgue measure when d is an integer. So the Hausdorff measure is the Lebesgue measure, simply scaled by a constant with $c_1 = 1$
- It is an outer measure on X . By Carathéodory's Extension Theorem, it is a measure on the σ -algebra of Hausdorff measurable sets
- Scaling property: For $\lambda > 0$, $\mathcal{H}^d(\lambda A) = \lambda^d \mathcal{H}^d(A)$ where $\lambda A := \{\lambda a : a \in A\}$

The following theorem is crucial to define the dimension of a set

Theorem

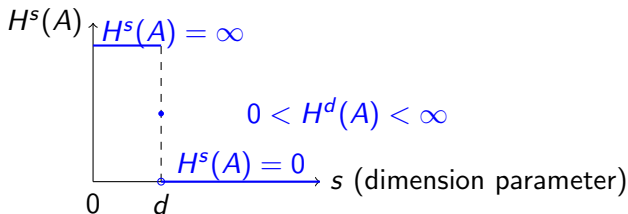
For $0 \leq s < t < \infty$ and $A \subseteq X$

- 1 $\mathcal{H}^s(A) < \infty \implies \mathcal{H}^t(A) = 0$
- 2 $\mathcal{H}^t(A) > 0 \implies \mathcal{H}^s(A) = \infty$

Threshold Behaviour

- This creates a "threshold behaviour" for any set A
- If $\mathcal{H}^d(A)$ is a finite, nonzero value, then for every $s \in [0, \infty)$ with $s \neq d$, we either have $\mathcal{H}^s(A)$ is infinite, or zero. More specifically,

$$\mathcal{H}^s(A) = \begin{cases} \infty & \text{if } s < d \\ \text{finite, nonzero number} & \text{if } s = d \\ 0 & \text{if } s > d \end{cases}$$



Hausdorff Dimension: Definition

Definition (Hausdorff dimension)

For a set $A \subseteq X$, the Hausdorff dimension is:

$$\dim(A) = \inf\{d \geq 0 : \mathcal{H}^d(A) = 0\} = \sup\{d \geq 0 : \mathcal{H}^d(A) = \infty\} \quad (3)$$

For this definition, we take $\sup \emptyset = 0$

- At the critical dimension $d = \dim(A)$, three cases are possible:
 - $\mathcal{H}^d(A) = 0$
 - $0 < \mathcal{H}^d(A) < \infty$
 - $\mathcal{H}^d(A) = \infty$
- The dimension can be infinity or zero

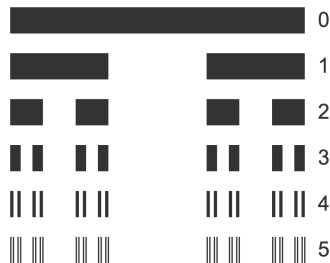
Properties of Hausdorff Dimension

For sets $A, B \subseteq X$:

- Monotonicity: If $A \subseteq B$, then $\dim(A) \leq \dim(B)$
- Countable stability: $\dim\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sup_{n \in \mathbb{N}} \dim(A_n)$
- For $\lambda > 0$, $\dim(\lambda A) = \dim(A)$
- It is possible for $\mathcal{H}^d(A) = \infty$ for every $d \in [0, \infty)$. An example is $\mathbb{R}$ are infinite dimensional in $(\mathbb{R}, d_D)$
- $\dim \mathbb{R}^n = n$ for the normal metric

Example: Cantor Set

- Begin with the interval $C_0 = [0, 1]$
- Remove the middle third $(\frac{1}{3}, \frac{2}{3})$ to get $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$
- Remove the middle third from each remaining interval to get $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$
- Continue this process indefinitely, with C_n the result at the n th iteration
- The Cantor set is the set as $n \rightarrow \infty$, that is, $\mathcal{C} := \lim_{n \rightarrow \infty} C_n = \bigcap_{n \in \mathbb{N}} C_n$



Cantor Set Properties

This Cantor Set is interesting because:

- It is uncountable
- It contains no intervals
- It has dimension $d = \frac{\ln 2}{\ln 3} \approx 0.631$ (this also means it has zero Lebesgue measure)
- $\mathcal{H}^d(\mathcal{C}) = 1$
- $\mathcal{H}_\delta^d(\mathcal{C}) < \mathcal{H}^d(\mathcal{C})$ for any $\delta > 0$

Thank you for your attention!

Please ask any questions